Discrete Probability Problems

Nine problems from ICPC-style contests, with solutions

These problems come from programming contests. Each statement keeps the probability core of the original and drops the programming layer (large inputs, modular output, data structures). Only discrete probability is used. Each problem asks one question that rests on one key idea; the solution names the idea first. The three problems whose idea is hardest to spot come with a hint. A remark after each solution collects extensions for discussion; they are not part of the problem. Problems 1–5 are the main set, in course order; Problems 6–9 are backups. The selection is explained in `prob_course_candidates.md`. Every numerical answer below was checked by exact enumeration, and each model was checked against the original problem's official sample outputs.

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Main problems

1 Mingle

Source: NAQ 2025, Problem G (QOJ). *Topic:* indicator variables.

Problem. Let $k \geq 1$ and $n \geq 2k+1$. There are n players and n rooms, both labelled by $\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$. The rooms stand in a circle, so room $n-1$ is next to room 0. Player i walks into a room chosen uniformly at random from the $2k+1$ rooms $i-k, \ldots, i+k$ (indices mod n), independently of the other players. A player who ends up alone in a room is a *winner*. Find the expected number of winners.

Solution. Key idea: count rooms, not players.

A winner is alone in a room, and a room with exactly one player holds exactly one winner. So the number of winners is

$$W = \sum_{r \in \mathbb{Z}_n} X_r, \qquad X_r = \mathbf{1}[\text{room } r \text{ holds exactly one player}]\,.$$

Fix a room r. Player i can enter room r only if $r \in \{i-k, \ldots, i+k\}$, that is, $i \in \{r-k, \ldots, r+k\}$. These are $2k+1$ distinct players because $n \geq 2k+1$. Each of them enters room r with probability $\frac{1}{2k+1}$, independently of the others. So the number of players in room r is $N_r \sim \mathrm{Bin}\big(2k+1, \frac{1}{2k+1}\big)$, and

$$\mathbb{P}(X_r = 1) = \mathbb{P}(N_r = 1) = (2k+1) \cdot \frac{1}{2k+1}\Big(1 - \frac{1}{2k+1}\Big)^{2k} = \Big(\frac{2k}{2k+1}\Big)^{2k}.$$

The indicators X_r are dependent, but linearity of expectation does not need independence:

$$\mathbb{E}[W] = n\Big(\frac{2k}{2k+1}\Big)^{2k}.$$

For example, $n = 3$, $k = 1$ gives $3 \cdot (2/3)^2 = 4/3$. □

Remark. With $M = 2k+1$, $\mathbb{E}[W]/n = (1 - 1/M)^{M-1} \to e^{-1}$ as $k \to \infty$, because $\mathrm{Bin}(M, 1/M)$ tends to Poisson(1) and $\mathbb{P}(\text{Poisson}(1) = 1) = e^{-1}$. If every player instead chooses among all n rooms, the same argument gives the classical value $n(1 - 1/n)^{n-1}$.

2 Accidental Arithmetic

Source: NWERC 2025, Problem A (QOJ). *Topic:* symmetry, linearity of expectation.

Problem. A positive integer N has decimal digits $d_1 d_2 \cdots d_L$. You type the digits from left to right into a faulty calculator. Into each of the $L-1$ gaps between consecutive digits it inserts, independently, a "+" with probability α, a "−" with probability α, and nothing with probability $1 - 2\alpha$, where $0 < \alpha \leq \frac{1}{2}$. The resulting expression, such as $12 - 3 + 45$, is then evaluated; a term may begin with zeros, so $1 + 05 = 6$. Find the expected value of the result, and evaluate it for $N = 12345$ and $\alpha = 0.45$.

Solution. Key idea: every term after the first is equally likely to be added or subtracted, whatever its digits, so only the first term counts.

Call a gap a *sign gap* if it receives a sign, and let A be the (random) set of sign gaps. The set A cuts the digits into terms $F = T_0, T_1, \ldots, T_s$, and the value is

$$V = F + \sum_{t=1}^{s} \sigma_t T_t,$$

where $\sigma_t \in \{+1, -1\}$ is the sign in front of T_t. Given A, the terms are fixed numbers, and the signs are independent, each ± 1 with probability $\frac{1}{2}$. So $\mathbb{E}[V \mid A] = F$, and therefore $\mathbb{E}[V] = \mathbb{E}[F]$.

Let P_l be the number formed by the first l digits, and let G be the first sign gap, counting gaps from the left, with $G = L$ if there is none. Then $F = P_G$, $\mathbb{P}(G = l) = 2\alpha(1-2\alpha)^{l-1}$ for $l < L$, and $\mathbb{P}(G = L) = (1-2\alpha)^{L-1}$. Hence

$$\mathbb{E}[V] = \sum_{l=1}^{L-1} 2\alpha(1-2\alpha)^{l-1} P_l + (1-2\alpha)^{L-1} N.$$

For $N = 12345$ and $\alpha = 0.45$ we have $1 - 2\alpha = 0.1$, and

$$\begin{aligned}\mathbb{E}[V] &= 0.9 \cdot 1 + 0.09 \cdot 12 + 0.009 \cdot 123 + 0.0009 \cdot 1234 + 0.0001 \cdot 12345 \\ &= 0.9 + 1.08 + 1.107 + 1.1106 + 1.2345 = 5.4321.\end{aligned}$$

□

Remark. If "+" and "−" have different probabilities β and γ, the same conditioning gives $\mathbb{E}[V] = (1-\mu)\mathbb{E}[F] + \mu\mathbb{E}[S]$ with $\mu = \frac{\beta-\gamma}{\beta+\gamma}$, where S is the sum of all terms taken with plus signs; $\mathbb{E}[S]$ follows by linearity over digits.

3 502 Bad Gateway

Source: ICPC Asia EC Online 2024 (II), Problem L (QOJ). *Topic:* geometric distribution.

Problem. Fix an integer $T \geq 1$. A timer is set to a value drawn uniformly from $\{1, \ldots, T\}$ and counts down by 1 each second; you are done when it reaches 0. At the end of every second at which the timer is still positive, you may *reset* it: it is then set to a fresh uniform value from $\{1, \ldots, T\}$, independent of the past. You fix a threshold $c \in \{1, \ldots, T\}$: whenever the timer is set to a value larger than c, you wait one second and reset; otherwise you let it run down. Find the expected time until you are done as a function of c, and its minimum over c when $T = 10$.

Solution. Key idea: the number of resets is geometric.

Each draw is kept with probability c/T, independently of the other draws, so the number of rejected draws is geometric with mean $\frac{1-c/T}{c/T} = \frac{T-c}{c}$. A rejected draw is larger than $c \geq 1$, so the timer is still positive after one second, and the draw costs exactly one second. The kept draw is uniform on $\{1, \ldots, c\}$ and costs its value, with mean $\frac{c+1}{2}$. Hence the expected time is

$$g(c) = \frac{T-c}{c} + \frac{c+1}{2} = \frac{T}{c} + \frac{c-1}{2}.$$

Since $g(c+1) - g(c) = \frac{1}{2} - \frac{T}{c(c+1)}$, the function g decreases while $c(c+1) < 2T$ and does not decrease afterwards. For $T = 10$ we have $4 \cdot 5 = 20 = 2T$, so $c = 4$ and $c = 5$ tie:

$$g(4) = \frac{10}{4} + \frac{3}{2} = 4 = \frac{10}{5} + 2 = g(5).$$

The minimal expected time is 4, against $g(10) = 5.5$ for never resetting. □

Remark. No strategy beats the best threshold. Let V be the optimal expected time from a fresh draw; after any reset the future is a fresh copy of the problem, so it costs V again. With s seconds showing, riding the timer out costs s and resetting costs V, so it is optimal to reset exactly when $s > V$: a threshold rule. In general the best threshold is the least c with $c(c+1) \geq 2T$, about $\sqrt{2T}$, and the minimum is $\sqrt{2T} - \frac{1}{2} + O(T^{-1/2})$; for $T = 1000$ it is $\frac{398}{9}$ at $c = 45$.

4 Uniform Chemistry

Source: SWERC 2022–2023, Problem B (QOJ). *Topic:* indicator variables, Markov chains.

Problem. Fix integers $1 \le a < n$. A particle starts at a. At each step, if it is at $x < n$, it jumps to a point chosen uniformly from $\{x+1, \dots, n\}$, independently of the past. Find the expected number of steps until it reaches n.

Hint: for $a < j < n$, find the probability that the particle ever visits j.

Solution. Key idea: the first jump into $\{j, \dots, n\}$ lands uniformly on that set.

Each step lands on a new point, and the last step lands on n. So the number of steps is

$$T = 1 + \sum_{j=a+1}^{n-1} I_j, \qquad I_j = \mathbf{1}[\text{the particle visits } j]\,.$$

Fix $a < j < n$. The particle starts below j and ends at n, so exactly one of its jumps is the first to land in $\{j, \dots, n\}$, and it starts from some $x < j$. From a point $x < j$, a jump lands on j with probability $\frac{1}{n-x}$ and lands in $\{j, \dots, n\}$ with probability $\frac{n-j+1}{n-x}$, which is $n-j+1$ times larger. Summing over the point $x < j$ the jump starts from,

$$\mathbb{P}(I_j = 1) = \sum_{x<j} \mathbb{P}(\text{visit } x)\, \frac{1}{n-x}, \qquad 1 = \sum_{x<j} \mathbb{P}(\text{visit } x)\, \frac{n-j+1}{n-x},$$

so $\mathbb{P}(I_j = 1) = \frac{1}{n-j+1}$. By linearity,

$$\mathbb{E}[T] = 1 + \sum_{j=a+1}^{n-1} \frac{1}{n-j+1} = 1 + \frac{1}{2} + \cdots + \frac{1}{n-a} = H_{n-a},$$

the harmonic number $H_m = \sum_{i=1}^{m} \frac{1}{i}$. Linearity does not need the I_j to be independent. □

Remark. The I_j are in fact independent: the path $a \to s_1 \to \cdots \to s_r \to n$ has probability $\frac{1}{n-a} \prod_i \frac{1}{n-s_i}$, which factorises as a product of $\mathbb{P}(I_j = 1)$ and $\mathbb{P}(I_j = 0)$ terms. Hence, with $m = n - a$, $\mathrm{Var}(T) = H_m - \sum_{i \le m} \frac{1}{i^2}$, and $\mathbb{E}[z^T] = z(z+1)\cdots(z+m-1)/m!$, so T is distributed like the number of cycles of a random permutation of m elements. The contest problem asks which of several independent particles reaches n first; for three particles with $a = 1$, $n = 3$, each wins (ties included) with probability $\frac{5}{8}$.

5 Jinx or Jackpot

Source: NERC 2025, Problem J (QOJ). *Topic:* Bayes' rule, value of information.

Problem. A slot machine pays even odds: if you bet x, you win x with probability p (a *jackpot*) and lose x with probability $1 - p$ (a *jinx*). You do not know p: it is 0.7 or 0.3, each with probability $\frac{1}{2}$, and given p the rounds are independent. You start with \$1000 and play two rounds. In each round you may bet any amount from 0 up to your current balance; a bet of 0 still shows you whether the round was a jackpot. Find a strategy that maximises the expected final balance, and the maximal expected profit.

Hint: start with round 2. Find the best bet there after a jackpot and after a jinx.

Solution. Key idea: the expected final balance is linear in the bet, so the best bet is either 0 or everything.

First, Bayes' rule. After a jackpot,

$$\mathbb{P}(p = 0.7 \mid W) = \frac{\frac{1}{2} \cdot 0.7}{\frac{1}{2} \cdot 0.7 + \frac{1}{2} \cdot 0.3} = 0.7, \qquad \hat{p}_W = 0.7 \cdot 0.7 + 0.3 \cdot 0.3 = 0.58$$

is the probability that the next round is a jackpot. Likewise, after a jinx $\mathbb{P}(p = 0.7 \mid L) = 0.3$ and $\hat{p}_L = 0.42$. Before any round the probability is 0.5.

Round 2. With balance w and jackpot probability $\hat{p}$, a bet x gives expected final balance $w + x(2\hat{p} - 1)$, which is linear in x. After a jackpot $\hat{p}_W > \frac{1}{2}$, so go all-in and expect $1.16\,w$. After a jinx $\hat{p}_L < \frac{1}{2}$, so bet nothing and keep w.

Round 1. A bet x from \$1000 gives expected final balance

$$\tfrac{1}{2}(1000 + x) \cdot 1.16 + \tfrac{1}{2}(1000 - x) \cdot 1 = 1080 + 0.08\,x,$$

again linear in x and largest at $x = 1000$. So go all-in in round 1 and, after a jackpot, all-in again. The expected final balance is \$1160, a profit of \$160; watching round 1 with a bet of 0 earns only \$80.

Round 1 is a fair bet, yet staking everything on it is optimal. A jackpot both doubles the balance and makes the machine look better, so the balance is largest exactly when the next bet is most favourable. □

Remark. The same linearity works for any number of rounds and any prior over p: by induction the optimal value is the balance times a factor that depends only on the outcomes seen, and every bet is 0 or all-in. If $p > \frac{1}{2}$ is known, going all-in every round earns $1000\big((2p)^k - 1\big)$, for example \$1702.71 for $p = 0.61$ and $k = 5$.

Backup problems

6 Orchard Fence

Source: 2025 ICPC Southeast USA Regional, Division 1, Problem A; listed under Mid-Atlantic USA in `data/tagged.json` (QOJ). *Topic:* linearity of expectation, counting.

Problem. The n vertices of a regular n-gon lie on a circle of radius 1. Keep k of them, chosen uniformly at random, where $3 \le k \le n$, and let L be the perimeter of the convex polygon they form. Find $\mathbb{E}[L]$, and evaluate it for $n = 6$, $k = 4$.

Hint: when is the chord from vertex i to vertex $i + s$ a side of the polygon?

Solution. Key idea: write L as a sum over chords and find the probability that each chord is a side.

Label the vertices $0, \dots, n-1$ counterclockwise. The sides of the kept polygon join cyclically consecutive kept vertices. For a vertex i and $1 \le s \le n-1$, the chord from i to $i + s$ is a side, traversed counterclockwise, exactly when i and $i + s$ are kept and none of the $s - 1$ vertices $i + 1, \dots, i + s - 1$ is kept. The other $k - 2$ kept vertices then come from the remaining $n - 1 - s$, so this happens for $\binom{n-1-s}{k-2}$ of the $\binom{n}{k}$ subsets. The chord has length $2 \sin \frac{\pi s}{n}$. By linearity,

$$\mathbb{E}[L] = \frac{n}{\binom{n}{k}} \sum_{s=1}^{n-1} 2 \sin \frac{\pi s}{n} \binom{n-1-s}{k-2},$$

where $\binom{m}{k-2} = 0$ for $m < k - 2$. For $n = 6$ and $k = 4$ the terms $s = 1, 2, 3$ are $1 \cdot \binom{4}{2} = 6$, $\sqrt{3} \cdot \binom{3}{2} = 3\sqrt{3}$ and $2 \cdot \binom{2}{2} = 2$, and the rest vanish, so

$$\mathbb{E}[L] = \frac{6}{15}\big(8 + 3\sqrt{3}\big) = \frac{16 + 6\sqrt{3}}{5} \approx 5.278.$$ □

Remark. For n points in general position, not necessarily convex, the same argument works: $z_i \to z_j$ is a counterclockwise hull edge exactly when every other kept point lies to its left. This gives $\mathbb{E}[L] = \sum_{i \neq j} |z_i z_j| \binom{\ell_{ij}}{k-2} / \binom{n}{k}$, where ℓ_{ij} counts the points to the left of $z_i \to z_j$; this is the contest solution. In the original the trees are disks of diameter d, and the fence has length $L + \pi d$.

7 Chopsticks

Source: 2025 ICPC Asia Taiwan Online Programming Contest, Problem H (QOJ). *Topic:* parity, linearity of expectation.

Problem. A box holds 12 chopsticks: 4 red, 4 blue and 4 green. You draw 6 of them uniformly at random and hand them out to 3 guests in pairs, making as many single-colour pairs as possible. Find the expected number of guests who get a mixed pair.

Solution. Key idea: the number of mixed pairs is half the number of colours drawn an odd number of times.

Let X_c be the number of drawn chopsticks of colour c. Colour c gives at most $\lfloor X_c/2 \rfloor$ single-colour pairs, and pairing within colours achieves $\sum_c \lfloor X_c/2 \rfloor$. The leftovers, one of each colour with X_c odd, are then paired arbitrarily. If O colours have an odd count, the single-colour pairs number $(6-O)/2$, so the number of mixed pairs is $M = 3 - (6-O)/2 = O/2$. By linearity and symmetry,

$$\mathbb{E}[M] = \frac{1}{2} \sum_c \mathbb{P}(X_c \text{ odd}) = \frac{3}{2}\, \mathbb{P}(X_{\text{red}} \text{ odd}).$$

The red count is hypergeometric, so

$$\mathbb{P}(X_{\text{red}} \text{ odd}) = \frac{\binom{4}{1}\binom{8}{5} + \binom{4}{3}\binom{8}{3}}{\binom{12}{6}} = \frac{224 + 224}{924} = \frac{16}{33}, \qquad \mathbb{E}[M] = \frac{3}{2} \cdot \frac{16}{33} = \frac{8}{11} \approx 0.727. \qquad \square$$

Remark. The same argument gives $\mathbb{E}[M] = \frac{1}{2} \sum_i \mathbb{P}(X_i \text{ odd})$ in general, with k_i chopsticks of type i, s in total and $2n$ drawn. For large inputs, write $\mathbb{P}(X \text{ odd})$ as $\frac{1}{2}\big(1 - \mathbb{E}[(-1)^X]\big)$ and use

$$\binom{s}{2n}\, \mathbb{E}\big[(-1)^{X_i}\big] = [z^{2n}]\, (1-z)^{k_i} (1+z)^{s-k_i},$$

where $[z^N]f$ is the coefficient of z^N in f. For 8 red and 8 blue with 10 drawn this is $[z^{10}](1-z^2)^8$ and gives $\mathbb{E}[M] = \frac{72}{143}$. If the $2n$ chopsticks are drawn independently, each of type i with probability p_i, then $\mathbb{E}[M] = \frac{1}{4} \sum_i \big(1 - (1-2p_i)^{2n}\big)$.

8 Crashing Competition Computer

Source: NWERC 2022, Problem C (QOJ). *Topic:* recursions for expectations.

Problem. You must type a text of 3 characters, one time unit per character. After each typed character, independently, the computer crashes with probability $\frac{1}{2}$. A crash costs 2 time units of recovery, after which you continue from the last save; the start of the text counts as saved. A save takes 5 time units and never crashes, and the text must be saved after its last character. Find the minimal expected time to finish.

Solution. Key idea: after a crash you are back where you were right after the last save, so the expected time of a block satisfies a one-line recursion.

Because a crash returns you to the state right after the last save, a strategy amounts to choosing where to save. Write $p = \frac{1}{2}$ for the crash probability, $q = 1 - p$ and $r = 2$ for the

recovery time. Let E_k be the expected time, starting from a save, until k characters have been typed with no crash in between. Once $k-1$ characters are typed, you type one more. With probability p it crashes; you pay r and need E_k more in expectation. Hence

$$E_k = E_{k-1} + 1 + p\,(r + E_k), \qquad \text{so} \qquad E_k = \frac{E_{k-1} + 1 + pr}{q} = 2(E_{k-1} + 2),$$

with $E_0 = 0$. This gives $E_1 = 4$, $E_2 = 12$ and $E_3 = 28$. Each block also ends with a save costing 5. Compare the four ways to place saves:

- only at the end (block 3): $28 + 5 = 33$;
- after character 1 (blocks $1, 2$): $(4 + 5) + (12 + 5) = 26$;
- after character 2 (blocks $2, 1$): $(12 + 5) + (4 + 5) = 26$;
- after characters 1 and 2 (blocks $1, 1, 1$): $3 \cdot (4 + 5) = 27$.

The minimal expected time is 26. □

Remark. In general $E_k = \left(\frac{1}{p} + r\right)\left(q^{-k} - 1\right)$. This is convex in k, so for a fixed number of saves the best blocks have equal lengths (within one), and only the number of saves remains to choose. That is the contest solution for texts of up to 2000 characters.

9 Disbursement on Quarantine Policy

Source: ICPC Asia Taiwan 2024, Problem D (QOJ). *Topic:* linearity of expectation, complements.

Problem. Nine passengers sit in a 3×3 grid of seats, and each is infected independently with probability $\frac{1}{2}$. An infected passenger is quarantined for 4 days. A healthy passenger with at least one infected edge-neighbour (sharing a side) is quarantined for 2 days. A healthy passenger with no infected edge-neighbour but at least one infected corner-neighbour (sharing only a corner) is quarantined for 1 day. Everyone else is not quarantined. Find the expected total number of quarantine days.

Solution. Key idea: linearity over seats, with each seat computed through the complement "no neighbour is infected".

Take a seat with e edge-neighbours and c corner-neighbours. The events "the passenger is infected", "no edge-neighbour is infected" and "no corner-neighbour is infected" concern disjoint sets of seats, so they are independent, with probabilities $\frac{1}{2}$, 2^{-e} and 2^{-c}. The expected quarantine length of this passenger is therefore

$$4 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2}\left(1 - 2^{-e}\right) + 1 \cdot \frac{1}{2}\,2^{-e}\left(1 - 2^{-c}\right).$$

The grid has three kinds of seats:

- 4 corner seats ($e = 2$, $c = 1$): $2 + \frac{3}{4} + \frac{1}{16} = \frac{45}{16}$;
- 4 side seats ($e = 3$, $c = 2$): $2 + \frac{7}{8} + \frac{3}{64} = \frac{187}{64}$;
- 1 centre seat ($e = 4$, $c = 4$): $2 + \frac{15}{16} + \frac{15}{512} = \frac{1519}{512}$.

By linearity, the expected total is

$$4 \cdot \frac{45}{16} + 4 \cdot \frac{187}{64} + \frac{1519}{512} = \frac{5760 + 5984 + 1519}{512} = \frac{13263}{512} \approx 25.90.$$ □

Remark. In the original problem the grid is up to 100×100, the day counts $d_0 \geq d_1 \geq d_2$ are inputs, and some passengers are known to be infected or healthy. The same argument works with each seat's infection probability in $\{0, \frac{1}{2}, 1\}$.